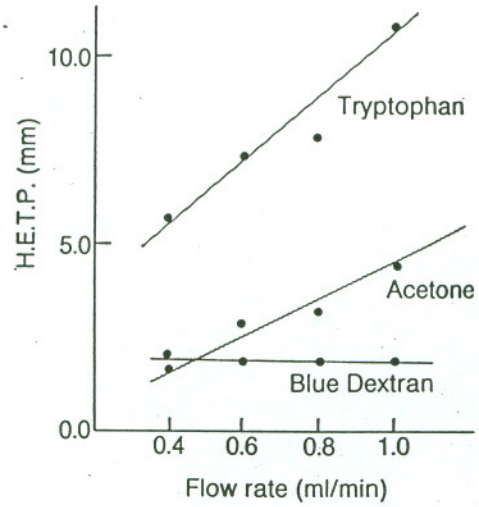
Analytical Cumulative Examination
April 26, 1997

1. (30 points) The data below obtained on a 0.4 x 50 cm Sephadex G-25 column (Anal. Chem., **52**, 910 (1980)) shows the dependence of the HETP on flow rate for the separation of three compounds by liquid chromatography.

   (a) Notice that the HETP is different for each compound. This is a typical observation for different compounds on the same column. Explain the basis for this observation using the Van Deemter equation.

   (b) The data for Blue Dextran are flat, but for tryptophan and acetone the HETP increases with increasing flow rate. Rationalize these observations in terms of the Van Deemter equation.



2. (30 points)

   (a) Consider a protein with a net negative charge at pH 8 which tightly adsorbs on an anion exchange column. How will a gradient of eluent pH from 8 to some lower value be useful for eluting the protein. Over the pH gradient the ionic strength stays constant. Explain your answer by providing detail on the nature of the adsorption interaction (solute/resin) and describe how the interaction is altered as a function of pH.

   (b) How would a gradient of ionic strength at constant pH be useful for eluting the protein? Explain your answer.

3. (40 points) The Departmental seminar program over the past year has provided many interesting and informative talks. For the four speakers given below, chose **one** and provide a brief overview of the speakers talk including an approximate title, detail summarizing the experimental work and discuss the important and/or surprising conclusions presented in the seminar:

   (a) Professor Kim Prather (UC-Riverside)

   (b) Professor Mario Rivera (Oklahoma State University)

   (c) Professor Grenville Turner (Manchester)

   (d) Professor Jerry Atwood (University of Missouri)